电子科技大学
University of Electronic Science and Technology of China

第2章 无约束高维优化

共轭法

共轭方向法

假设$f(\boldsymbol{x})$定义域内一阶可微，要求找出$f(\boldsymbol{x})$的局部极小点。即

$$\min_{\boldsymbol{x} \in \mathbb{R}^n} f(\boldsymbol{x})$$

我们介绍一种新的方法：共轭方向法。

- 特性：
  - 对于n维二次型问题，能够在n步之内得到结果
  - 不需要计算黑塞矩阵
  - 不需要对黑塞矩阵求逆

共轭方向法

- 一般来说，目标函数在极小点附近近似二次函数，因此，一个算法对于二次函数比较优秀，就可望对于一般函数也有较好的效果。
- 给出一个n变量的二次型函数：

$$f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} - \boldsymbol{x}^T\boldsymbol{b}$$

$\boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{Q} = \boldsymbol{Q}^T > 0$。

我们先探讨该二次型下的共轭方向法，然后推广到一般情况上。

共轭方向法

- **共轭定义**：设$\boldsymbol{Q} \in \mathbb{R}^{n\times n}$且$\boldsymbol{Q} = \boldsymbol{Q}^T$，，对于方向$\boldsymbol{d}^{(0)}, \boldsymbol{d}^{(1)}, \boldsymbol{d}^{(2)}, \dots, \boldsymbol{d}^{(\mathrm{m})}$，如果对于所有的$i \neq j$，都有$\boldsymbol{d}^{(i)T}\boldsymbol{Q}\boldsymbol{d}^{(j)} = 0$，则称它们是关于$\boldsymbol{Q}$**共轭**的。

共轭方向法

- **性质：** $\boldsymbol{Q}$为$n \times n$的对称正定矩阵，如果方向$\boldsymbol{d}^{(0)}, \boldsymbol{d}^{(1)}, \boldsymbol{d}^{(2)}, \dots, \boldsymbol{d}^{(k)} \in \mathbb{R}^n, k \leq n-1$非零，且是关于Q共轭的，那么它们是线性无关的。

证明：设存在一组$\alpha_0, \dots, \alpha_k$，满足

$$\alpha_0 \boldsymbol{d}^{(0)} + \alpha_1 \boldsymbol{d}^{(1)} + \cdots + \alpha_k \boldsymbol{d}^{(k)} = 0$$

等式两边同乘以$\boldsymbol{d}^{(\mathrm{j})^{\mathrm{T}}} \boldsymbol{Q} \Rightarrow$

$$\alpha_0 \boldsymbol{d}^{(\mathrm{j})^{\mathrm{T}}} \boldsymbol{Q} \boldsymbol{d}^{(0)} + \alpha_1 \boldsymbol{d}^{(\mathrm{j})^{\mathrm{T}}} \boldsymbol{Q} \boldsymbol{d}^{(1)} + \cdots + \alpha_k \boldsymbol{d}^{(\mathrm{j})^{\mathrm{T}}} \boldsymbol{Q} \boldsymbol{d}^{(k)} = 0$$

根据共轭定义$\Rightarrow$第i项$\boldsymbol{d}^{(j)^{\mathrm{T}}} \boldsymbol{Q} \boldsymbol{d}^{(\mathrm{i})} = 0, (i \neq j) \Rightarrow$等式变为

$$\alpha_j \boldsymbol{d}^{(j)^{\mathrm{T}}} \boldsymbol{Q} \boldsymbol{d}^{(j)} = 0$$

由于$\boldsymbol{Q}$对称正定$\Rightarrow \alpha_j = 0$。同理，$\alpha_0, \dots, \alpha_k = 0$。得证。

- **推论**：n维空间中互相共轭的非零向量不超过n个。

共轭方向法

- 对于

$$\min f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} - \boldsymbol{x}^T\boldsymbol{b}, \boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{Q} = \boldsymbol{Q}^T > 0$$

给定初始点$\boldsymbol{x}^{(0)}$和一组关于$\boldsymbol{Q}$共轭的方向$\boldsymbol{d}^{(0)}, \boldsymbol{d}^{(1)}, \boldsymbol{d}^{(2)}, \dots, \boldsymbol{d}^{(\mathrm{n}-1)}$，共轭方向法的仍旧是一种迭代法：

1. 选定$\boldsymbol{x}^0, k = 0$
2. 在第k步时，计算$\boldsymbol{x}^{(\mathrm{k}+1)} = \boldsymbol{x}^{(k)} + \alpha_k \boldsymbol{d}^{(k)}$，其中$\alpha_k = -\frac{\boldsymbol{g}^{(k)T}\boldsymbol{d}^{(k)}}{\boldsymbol{d}^{(k)T}\boldsymbol{Q}\boldsymbol{d}^{(k)}}, \boldsymbol{g}^{(k)} = \nabla f\left(\boldsymbol{x}^{(k)}\right) = \boldsymbol{Q}\boldsymbol{x}^{(k)} - \boldsymbol{b}$。
3. 如果$k = n - 1$,返回$\boldsymbol{x}^{(n)}$,否则$k = k + 1$,返回第2步。

共轭方向法

- α_k利用了和最速下降法一样的规则选取，即$\alpha_k = argmin_\alpha f(\boldsymbol{x}^{(k)} + \alpha \boldsymbol{d}^{(k)})$。

令$\phi(\alpha) = f(\boldsymbol{x}^{(k)} + \alpha \boldsymbol{d}^{(k)})$

则$\phi'(\alpha) =$

$$\nabla f\left(\boldsymbol{x}^{(k)} + \alpha \boldsymbol{d}^{(k)}\right)^T \boldsymbol{d}^{(k)} = \boldsymbol{d}^{(k)T}(\boldsymbol{Q}\boldsymbol{x}^{(k)} + \alpha \boldsymbol{Q}\boldsymbol{d}^{(k)} - \boldsymbol{b})$$

由一阶必要条件$\phi'(\alpha) = 0$得，

$$\alpha = -\frac{\boldsymbol{d}^{(k)T}\left(\boldsymbol{Q}\boldsymbol{x}^{(k)} - \boldsymbol{b}\right)}{\boldsymbol{d}^{(k)T}\boldsymbol{Q}\boldsymbol{d}^{(k)}} = -\frac{\boldsymbol{g}^{(k)T}\boldsymbol{d}^{(k)}}{\boldsymbol{d}^{(k)T}\boldsymbol{Q}\boldsymbol{d}^{(k)}}$$

共轭方向法

- **共轭方向法的收敛性质**：$f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} - \boldsymbol{x}^T\boldsymbol{b}, \boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{Q} = \boldsymbol{Q}^T > 0$，共轭方向法能在$n$次收敛到全局唯一的极小点$\boldsymbol{x}^*$,即$\boldsymbol{x}^{(n)} = \boldsymbol{x}^*$。

证明：由于方向$\boldsymbol{d}^{(i)}$，$i = 0,1,\dots,n-1$ 线性无关，存在β满足：

$$\boldsymbol{x}^* - \boldsymbol{x}^{(0)} = \beta_0\boldsymbol{d}^{(0)} + \cdots + \beta_{n-1}\boldsymbol{d}^{(n-1)}$$

等号两端同时左乘$\boldsymbol{d}^{(k)T}\boldsymbol{Q}$，$0 \le k < n$，由共轭性质$\boldsymbol{d}^{(k)T}\boldsymbol{Q}\boldsymbol{d}^{(i)} = 0, k \ne i$，有

$$\boldsymbol{d}^{(k)T}\boldsymbol{Q}\left(\boldsymbol{x}^* - \boldsymbol{x}^{(\mathbf{0})}\right) = \beta_k\boldsymbol{d}^{(k)T}\boldsymbol{Q}\boldsymbol{d}^{(k)}$$

$$\Rightarrow \beta_k = \frac{\boldsymbol{d}^{(k)T}\boldsymbol{Q}\left(\boldsymbol{x}^* - \boldsymbol{x}^{(\mathbf{0})}\right)}{\boldsymbol{d}^{(k)T}\boldsymbol{Q}\boldsymbol{d}^{(k)}}$$

另一方面，根据算法：$\boldsymbol{x}^{(n)} = \boldsymbol{x}^{(0)} + \alpha_0\boldsymbol{d}^{(0)} + \cdots + \alpha_{n-1}\boldsymbol{d}^{(n-1)}$。

所以，如果能证明$\alpha_k = \beta_k$，就可以完成证明。

共轭方向法

证明（续）：

$$\boldsymbol{x}^{(k)} = \boldsymbol{x}^{(0)} + \alpha_0 \boldsymbol{d}^{(0)} + \cdots + \alpha_{k-1} \boldsymbol{d}^{(k-1)} \ \forall 0 \le k < n$$

$$\Rightarrow \boldsymbol{x}^{(k)} - \boldsymbol{x}^{(0)} = \alpha_0 \boldsymbol{d}^{(0)} + \cdots + \alpha_{k-1} \boldsymbol{d}^{(k-1)}$$

而$\boldsymbol{x}^* - \boldsymbol{x}^{(0)} = \left(\boldsymbol{x}^* - \boldsymbol{x}^{(k)}\right) + \left(\boldsymbol{x}^{(k)} - \boldsymbol{x}^{(0)}\right) = \left(\boldsymbol{x}^* - \boldsymbol{x}^{(k)}\right) + \alpha_0 \boldsymbol{d}^{(0)} + \cdots + \alpha_{k-1} \boldsymbol{d}^{(k-1)}$

等式两端同时乘以$\boldsymbol{d}^{(k)T}\boldsymbol{Q}$，

$$\begin{aligned}
&\boldsymbol{d}^{(k)T}\boldsymbol{Q}\left(\boldsymbol{x}^* - \boldsymbol{x}^{(\boldsymbol{0})}\right) \\
&= \boldsymbol{d}^{(k)T}\boldsymbol{Q}\left(\boldsymbol{x}^* - \boldsymbol{x}^{(k)}\right) + \alpha_0 \boldsymbol{d}^{(k)T}\boldsymbol{Q}\boldsymbol{d}^{(0)} + \cdots + \alpha_{k-1} \boldsymbol{d}^{(k)T}\boldsymbol{Q}\boldsymbol{d}^{(k-1)} \\
&= \boldsymbol{d}^{(k)\boldsymbol{T}}\boldsymbol{Q}\left(\boldsymbol{x}^* - \boldsymbol{x}^{(k)}\right) = \boldsymbol{d}^{(k)\boldsymbol{T}}\left(\boldsymbol{Q}\boldsymbol{x}^* - \boldsymbol{Q}\boldsymbol{x}^{(k)}\right)
\end{aligned}$$

另一方面$\boldsymbol{Q}\boldsymbol{x}^* - \boldsymbol{b} = 0$，$\boldsymbol{Q}\boldsymbol{x}^{(k)} - \boldsymbol{b} = \boldsymbol{g}^{(k)}$，所以

$$\boldsymbol{d}^{(k)\boldsymbol{T}}\left(\boldsymbol{Q}\boldsymbol{x}^* - \boldsymbol{Q}\boldsymbol{x}^{(k)}\right) = \boldsymbol{d}^{(k)T}\left(\boldsymbol{b} - \boldsymbol{b} - \boldsymbol{g}^{(k)}\right) = -\boldsymbol{d}^{(k)T}\boldsymbol{g}^{(k)}$$

这说明$\beta_k = \frac{\boldsymbol{d}^{(k)T}\boldsymbol{Q}\left(\boldsymbol{x}^* - \boldsymbol{x}^{(\boldsymbol{0})}\right)}{\boldsymbol{d}^{(k)T}\boldsymbol{Q}\boldsymbol{d}^{(k)}} = \alpha_k$，证毕。

共轭方向法

- 例：利用基本的共轭方向法求函数

$$f(x_1,x_2)=\frac{1}{2}\boldsymbol{x}^T\begin{bmatrix}4 & 2\\ 2 & 2\end{bmatrix}\boldsymbol{x}-\boldsymbol{x}^T\begin{bmatrix}-1\\ 1\end{bmatrix},\boldsymbol{x}\in\mathbb{R}^2$$

的极小点，初始点为$x^{(0)}=[0,0]^T$，给定的共轭方向 $d^{(0)}=[1,0]^T$，$\boldsymbol{d}^{(1)}=[-\frac{3}{8},\frac{3}{4}]^T$

解：$k=0$:

$$\boldsymbol{g}^{(0)}=-\boldsymbol{b}=[1,-1]^T,$$

$$\alpha_0=-\frac{\boldsymbol{g}^{(\boldsymbol{0})\boldsymbol{T}}\boldsymbol{d}^{(0)}}{\boldsymbol{d}^{(0)T}\boldsymbol{Q}\boldsymbol{d}^{(0)}}=-\frac{[1\ \ -1]\begin{bmatrix}1\\ 0\end{bmatrix}}{[1\quad 0]\begin{bmatrix}4 & 2\\ 2 & 2\end{bmatrix}\begin{bmatrix}1\\ 0\end{bmatrix}}=-\frac{1}{4}.$$

$$\boldsymbol{x}^{(1)}=\boldsymbol{x}^{(0)}+\alpha_0\boldsymbol{d}^{(0)}=\begin{bmatrix}0\\ 0\end{bmatrix}-\frac{1}{4}\begin{bmatrix}1\\ 0\end{bmatrix}=\begin{bmatrix}-\frac{1}{4}\\ 0\end{bmatrix}$$

共轭方向法

- 例：利用基本的共轭方向法求函数

$$f(x_1,x_2)=\frac{1}{2}\boldsymbol{x}^T\begin{bmatrix}4 & 2\\ 2 & 2\end{bmatrix}\boldsymbol{x}-\boldsymbol{x}^T\begin{bmatrix}-1\\ 1\end{bmatrix},\boldsymbol{x}\in\mathbb{R}^2$$

的极小点，初始点为$\boldsymbol{x}^{(0)}=[0,0]^T$，给定的共轭方向 $\boldsymbol{d}^{(0)}=[1,0]^T$，$\boldsymbol{d}^{(1)}=[-\frac{3}{8},\frac{3}{4}]^T$

解：$k=1$:

$$\boldsymbol{g}^{(1)}=\boldsymbol{Q}\boldsymbol{x}^{(1)}-\boldsymbol{b}=\begin{bmatrix}4 & 2\\ 2 & 2\end{bmatrix}\begin{bmatrix}-\frac{1}{4}\\ 0\end{bmatrix}-\begin{bmatrix}-1\\ 1\end{bmatrix}=\begin{bmatrix}0\\ -\frac{3}{2}\end{bmatrix}$$

$$\alpha_1=-\frac{\boldsymbol{g}^{(1)T}\boldsymbol{d}^{(1)}}{\boldsymbol{d}^{(1)T}\boldsymbol{Q}\boldsymbol{d}^{(1)}}=-\frac{\begin{bmatrix}0 & -\frac{3}{2}\end{bmatrix}\begin{bmatrix}-\frac{3}{8}\\ \frac{3}{4}\end{bmatrix}}{\begin{bmatrix}-\frac{3}{8} & \frac{3}{4}\end{bmatrix}\begin{bmatrix}4 & 2\\ 2 & 2\end{bmatrix}\begin{bmatrix}-\frac{3}{8}\\ \frac{3}{4}\end{bmatrix}}=2$$

$$\boldsymbol{x}^{(2)}=\boldsymbol{x}^{(1)}+\alpha_1\boldsymbol{d}^{(1)}=\begin{bmatrix}-\frac{1}{4}\\ 0\end{bmatrix}+2\begin{bmatrix}-\frac{3}{8}\\ \frac{3}{4}\end{bmatrix}=\begin{bmatrix}-1\\ \frac{3}{2}\end{bmatrix}$$

共轭方向法

- 在共轭方向法中，对所有的k，$0 \le k \le n$，$0 \le i \le k$，都有$\boldsymbol{g}^{(k+1)T}\boldsymbol{d}^{(i)} = 0$。

证明：归纳法，

(1)$\boldsymbol{g}^{(1)T}\boldsymbol{d}^{(0)} = 0$,

(2)假设$\boldsymbol{g}^{(k)T}\boldsymbol{d}^{(i)} = 0$对于$i \le k-1$成立，证明$\boldsymbol{g}^{(k+1)T}\boldsymbol{d}^{(i)} = 0$对于$i \le k$也成立。

- 注意，我们已经证明过：$\boldsymbol{g}^{(k+1)T}\boldsymbol{d}^{(k)} = 0$对于以$\boldsymbol{d}^{(k)}$为下降方向的最速下降法均成立。

共轭方向法

- 现在的问题是：怎样找出一组Q共轭的向量?

- 我们首先简要介绍Gram – Schmidt向量组共轭方法，该方法从一组已知的线性无关向量组构造共轭向量。

Gram – Schmidt

- $\boldsymbol{p}^{(0)},\dots,\boldsymbol{p}^{(n-1)}$是一组线性无关向量组（$\boldsymbol{p}^{(0)}$ 为下降方向），首先取$\boldsymbol{d}^{(0)}=\boldsymbol{p}^{(0)}$。设已经求得$\boldsymbol{d}^{(0)},\dots,\boldsymbol{d}^{(k)}$, 现在构造$\boldsymbol{d}^{(k+1)}$

$$\boldsymbol{d}^{(k+1)}=\boldsymbol{p}^{(k+1)}+\sum_{i=0}^{k}\beta_{k+1,i}\,\boldsymbol{d}^{(i)}$$

$\beta_{k+1,i}$是待定系数。为了使$\boldsymbol{d}^{(k+1)}$和$\boldsymbol{d}^{(0)},\dots,\boldsymbol{d}^{(k)}$关于$\boldsymbol{Q}$共轭，应有

$$\left(\boldsymbol{d}^{(r)}\right)^T\boldsymbol{Q}\boldsymbol{d}^{(k+1)}=\left(\boldsymbol{d}^{(r)}\right)^T\boldsymbol{Q}\boldsymbol{p}^{(k+1)}+\sum_{i=0}^{k}\beta_{k+1,r}\left(\boldsymbol{d}^{(r)}\right)^T\boldsymbol{Q}\boldsymbol{d}^{(i)}=0$$

对于所有$0\le r\le k$成立。

Gram − Schmidt

对于$i = r$时，我们得到

$$\beta_{k+1,i} = -\frac{\left(\boldsymbol{d}^{(i)}\right)^T \boldsymbol{Q}\boldsymbol{p}^{(k+1)}}{(\boldsymbol{d}^{(i)})^T \boldsymbol{Q}\boldsymbol{d}^{(i)}}$$

于是，

$$\boldsymbol{d}^{(k+1)} = \boldsymbol{p}^{(k+1)} - \sum_{i=0}^{k} \frac{\left(\boldsymbol{d}^{(i)}\right)^T \boldsymbol{Q}\boldsymbol{p}^{(k+1)}}{(\boldsymbol{d}^{(i)})^T \boldsymbol{Q}\boldsymbol{d}^{(i)}} \boldsymbol{d}^{(i)}$$

Gram − Schmidt

- 性质（习题10.1）：给出$f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} - \boldsymbol{x}^T\boldsymbol{b}, \boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{Q} = \boldsymbol{Q}^T > 0$， $\boldsymbol{p}^{(0)}, \dots, \boldsymbol{p}^{(n-1)}$是一组线性无关向量组，使用Gram − Schmidt构造出的向量$\boldsymbol{d}^{(0)}, \dots, \boldsymbol{d}^{(n-1)}$是关于$Q$的共轭方向。

证明：上述推导过程可得。

共轭梯度法

- 我们接下来介绍一种更加高效的方法：共轭梯度法。

- 共轭梯度法每一次利用**上一次搜索方向**和**目标函数在当前迭代点的梯度向量**之间的线性组合构造一个新方向，即$\boldsymbol{d}^{(k+1)} = -\boldsymbol{g}^{(k+1)} + \beta_k \boldsymbol{d}^{(k)}$，使其与前面已经产生搜索方向组成$\boldsymbol{Q}$共轭方向。

共轭梯度法

- $\mathrm{k}=0$时：

$$\boldsymbol{d}^{(0)}=-\nabla f\left(\boldsymbol{x}^{(0)}\right)=-\boldsymbol{g}^{(0)}$$

那么$\boldsymbol{x}^{(1)}=\boldsymbol{x}^{(0)}+\alpha_0\boldsymbol{d}^{(0)}$，其中$\alpha_0=-\frac{\boldsymbol{g}^{(0)T}\boldsymbol{d}^{(0)}}{\boldsymbol{d}^{(0)T}\boldsymbol{Q}\boldsymbol{d}^{(0)}}$

- $\mathrm{k}\geq 1$时：设第k次的方向为$\boldsymbol{d}^{(k)}$ $(k=1,\ldots,\mathrm{n}-1)$，第$k-1$次的梯度是$\boldsymbol{g}^{(k-1)}$，设

$$\boldsymbol{d}^{(k)}=-\boldsymbol{g}^{(k)}+\beta_{k-1}\boldsymbol{d}^{(k-1)}$$

为了使$\boldsymbol{d}^{(k)}$和$\boldsymbol{d}^{(k-1)}$关于$\boldsymbol{Q}$共轭，应有

$$\left(\boldsymbol{d}^{(k)}\right)^T\boldsymbol{Q}\boldsymbol{d}^{(k-1)}=-\left(\boldsymbol{g}^{(k)}\right)^T\boldsymbol{Q}\boldsymbol{d}^{(k-1)}+\beta_{k-1}\left(\boldsymbol{d}^{(k-1)}\right)^T\boldsymbol{Q}\boldsymbol{d}^{(k-1)}=0$$

因此

$$\beta_{k-1}=\frac{\left(\boldsymbol{g}^{(k)}\right)^T\boldsymbol{Q}\boldsymbol{d}^{(k-1)}}{(\boldsymbol{d}^{(k-1)})^T\boldsymbol{Q}d^{(k-1)}}$$

共轭梯度法

- 性质：给出$f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} - \boldsymbol{x}^T\boldsymbol{b}, \boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{Q} = \boldsymbol{Q}^T > 0$，共轭梯度法的搜索方向$\boldsymbol{d}^{(0)}, \boldsymbol{d}^{(1)}, \boldsymbol{d}^{(2)}, \dots, \boldsymbol{d}^{(k)}$是关于$\boldsymbol{Q}$的共轭方向。

证明：数学归纳法。参考P126,10.1

共轭梯度法

1. $k=0$，选择初始点$\boldsymbol{x}^{(0)}$
2. 计算$\boldsymbol{g}^{(0)}=\nabla f\left(x^{(0)}\right)$。如果满足停机条件（如$\left\|\boldsymbol{g}^{(0)}\right\| \leq \varepsilon$），则停止。否则设$\boldsymbol{d}^{(0)}=-\boldsymbol{g}^{(0)}$.
3. 计算$\alpha_k=-\frac{\left(\boldsymbol{g}^{(k)}\right)^T \boldsymbol{d}^{(k)}}{\left(\boldsymbol{d}^{(k)}\right)^T \boldsymbol{Q} d^{(k)}}$
4. 计算$\boldsymbol{x}^{(k+1)}=\boldsymbol{x}^{(k)}+\alpha_k \boldsymbol{d}^{(k)}$
5. 计算$\boldsymbol{g}^{(k+1)}=\nabla f(\boldsymbol{x}^{(k+1)})$,检查停止条件。如满足，返回$\boldsymbol{x}^{(k+1)}$；否则进入下一步
6. 计算$\beta_k=\frac{\left(\boldsymbol{g}^{(k+1)}\right)^T \boldsymbol{Q} \boldsymbol{d}^{(k)}}{\left(\boldsymbol{d}^{(k)}\right)^T \boldsymbol{Q} \boldsymbol{d}^{(k)}}$
7. 计算$\boldsymbol{d}^{(k+1)}=-\boldsymbol{g}^{(k+1)}+\beta_k \boldsymbol{d}^{(k)}$
8. 令$k=k+1$,返回第三步

共轭梯度法

例：利用共轭梯度法求

$$\min f(x_1,x_2,x_3)=\frac{3}{2}x_1^2+2x_2^2+\frac{3}{2}x_3^2+x_1x_3+2x_2x_3-3x_1-x_3.$$

初始点$x^{(0)}=[0,0,0]^T$

解:$f(\boldsymbol{x})=\frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x}-\boldsymbol{x}^T\boldsymbol{b},\boldsymbol{Q}=\begin{bmatrix}3&0&1\\0&4&2\\1&2&3\end{bmatrix}\boldsymbol{b}=\begin{bmatrix}3\\0\\1\end{bmatrix}$

$$g(\boldsymbol{x})=\nabla f(\boldsymbol{x})=\boldsymbol{Q}\boldsymbol{x}-\boldsymbol{b}=[3x_1+x_3-3,4x_2+2x_3,x_1+2x_2+3x_3-1]^T$$

- $\mathrm{k}=0$： $\boldsymbol{g}^{(\mathbf{0})}=[-3,\ 0,-1]^T$,

$$\boldsymbol{d}^{(\mathbf{0})}=-\boldsymbol{g}^{(0)},$$

$$\alpha_0=-\frac{\boldsymbol{g}^{(0)T}\boldsymbol{d}^{(\mathbf{0})}}{\boldsymbol{d}^{(\mathbf{0})\boldsymbol{T}}\boldsymbol{Q}\boldsymbol{d}^{(0)}}=\frac{10}{36}=0.$$

$$\boldsymbol{x}^{(\mathbf{1})}=\boldsymbol{x}^{(\mathbf{0})}+\alpha_0\boldsymbol{d}^{(\mathbf{0})}=[0.8333,0,0.2778]^T.$$

$$\boldsymbol{d}^{(1)}=-\boldsymbol{g}^{(1)}+\beta_0\boldsymbol{d}^{(0)}=[0.4630,-0.5556,-0.5864]^T.$$

共轭梯度法

- $k=1$： $\alpha_1=-\frac{\boldsymbol{g}^{(1)T}\boldsymbol{d}^{(1)}}{\boldsymbol{d}^{(1)T}\boldsymbol{Q}\boldsymbol{d}^{(1)}}=0.2187$

$$\boldsymbol{x}^{(2)}=\boldsymbol{x}^{(1)}+\alpha_1\boldsymbol{d}^{(1)}=[0.9346,-0.1215,0.1495]^T$$

$$\boldsymbol{g}^{(2)}=\nabla f(\boldsymbol{x}^{(2)})=[-0.04673,-0.1869,0.1402]^T$$

$$\beta_1=\frac{\boldsymbol{g}^{(2)T}\boldsymbol{Q}\boldsymbol{d}^{(1)}}{\boldsymbol{d}^{(1)T}\boldsymbol{Q}\boldsymbol{d}^{(1)}}=0.07075$$

$$\boldsymbol{d}^{(2)}=-\boldsymbol{g}^{(2)}+\beta_1\boldsymbol{d}^{(1)}=[0.07948,0.1476,-0.1817]^T$$

- $k=2$： $\alpha_2=-\frac{\boldsymbol{g}^{(2)T}\boldsymbol{d}^{(2)}}{\boldsymbol{d}^{(2)T}\boldsymbol{Q}\boldsymbol{d}^{(2)}}=0.8231,$

$$\boldsymbol{x}^{(3)}=\boldsymbol{x}^{(2)}+\alpha_2\boldsymbol{d}^{(2)}=[1.000,0.000,0.000]^T$$

$$\boldsymbol{g}^{(3)}=\nabla f(\boldsymbol{x}^{(3)})=0$$

目标函数f是一个三变量的二次型函数，如预料，三次迭代得到了极小点$\boldsymbol{x}^*=\boldsymbol{x}^{(3)}$

非二次型问题

- 对于非二次型，显然可以利用泰勒展开式的二阶近似，将共轭梯度法直接推广到一般的二阶连续可微函数中去。$F(x)$对应于共轭梯度法的$\boldsymbol{Q}$矩阵。

- 但是,该方法要求函数可微，且每次迭代都需要计算黑塞矩阵，运算量可能很大

非二次型问题

- 观察到Q矩阵只出现在求α_k和β_k的公式中。我们可以进行以下处理

1. 将步长$\alpha_k = argmin_\alpha f(\boldsymbol{x}^{(k)} + \alpha \boldsymbol{d}^{(k)})$的搜索方式变为一维搜索算法。
2. 将β_k的计算改为与黑塞矩阵无关。方法有：

- Hestenes-Stiefel公式：$\beta_k = \frac{\boldsymbol{g}^{(k+1)T}[\boldsymbol{g}^{(k+1)} - \boldsymbol{g}^{(k)}]}{\boldsymbol{d}^{(k)T}\boldsymbol{g}^{(k+1)} - \boldsymbol{d}^{(k)T}\boldsymbol{g}^{(k)}}$
- Polak –Ribière公式：$\beta_k = \frac{\boldsymbol{g}^{(k+1)T}[\boldsymbol{g}^{(k+1)} - \boldsymbol{g}^{(k)}]}{\boldsymbol{g}^{(k)T}\boldsymbol{g}^{(k)}}$
- Fletcher-Reeves公式：$\beta_k = \frac{\boldsymbol{g}^{(k+1)T}\boldsymbol{g}^{(k+1)}}{\boldsymbol{g}^{(k)T}\boldsymbol{g}^{(k)}}$

非二次型问题

- **Hestenes-Stiefel公式：** $\beta_k = \frac{\boldsymbol{g}^{(k+1)T}[\boldsymbol{g}^{(k+1)}-\boldsymbol{g}^{(k)}]}{\boldsymbol{d}^{(k)T}[\boldsymbol{g}^{(k+1)}-\boldsymbol{g}^{(k)}]}$

该公式使用$\frac{\boldsymbol{g}^{(k+1)}-\boldsymbol{g}^{(k)}}{\alpha_k}$替换原公式的 $\boldsymbol{Q}\boldsymbol{d}^{(k)}$，对于二次型函数这两项是相等的。

已知$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \alpha_k \boldsymbol{d}^{(k)}$,两边同乘以$\boldsymbol{Q}$，然后同时减去$\boldsymbol{b}$得到

$$\boldsymbol{g}^{(k+1)} = \boldsymbol{g}^{(k)} + \alpha_k \boldsymbol{Q}\boldsymbol{d}^{(k)}$$

$$\Rightarrow \boldsymbol{Q}\boldsymbol{d}^{(k)} = \frac{\boldsymbol{g}^{(k+1)} - \boldsymbol{g}^{(k)}}{\alpha_k}$$

带入到$\beta_k = \frac{\left(\boldsymbol{g}^{(k+1)}\right)^T \boldsymbol{Q}\boldsymbol{d}^{(k)}}{\left(\boldsymbol{d}^{(k)}\right)^T \boldsymbol{Q}\boldsymbol{d}^{(k)}}$可得Hestenes-Stiefel公式

非二次型问题

- **Polak –Ribière公式：** $\beta_k = \frac{\boldsymbol{g}^{(k+1)T}[\boldsymbol{g}^{(k+1)} - \boldsymbol{g}^{(k)}]}{\boldsymbol{g}^{(k)T}\boldsymbol{g}^{(k)}}$

该公式通过以下思想得到：将Hestenes-Stiefel分母展开：

$$\beta_k = \frac{\boldsymbol{g}^{(k+1)T}[\boldsymbol{g}^{(k+1)} - \boldsymbol{g}^{(k)}]}{\boldsymbol{d}^{(k)T}\boldsymbol{g}^{(k+1)} - \boldsymbol{d}^{(k)T}\boldsymbol{g}^{(k)}}$$

利用共轭梯度法性质$\boldsymbol{g}^{(k+1)T}\boldsymbol{d}^{(k)} = 0 \Longrightarrow \beta_k = -\frac{\boldsymbol{g}^{(k+1)T}[\boldsymbol{g}^{(k+1)} - \boldsymbol{g}^{(k)}]}{\boldsymbol{d}^{(k)T}\boldsymbol{g}^{(k)}}$

进一步公式$\boldsymbol{d}^{(k)} = -\boldsymbol{g}^{(k)} + \beta_{k-1}\boldsymbol{d}^{(k-1)}$

$$\Longrightarrow \boldsymbol{g}^{(k)T}\boldsymbol{d}^{(k)} = -\boldsymbol{g}^{(k)T}\boldsymbol{g}^{(k)} + \beta_{k-1}\boldsymbol{g}^{(k)T}\boldsymbol{d}^{(k-1)} = -\boldsymbol{g}^{(k)T}\boldsymbol{g}^{(k)}$$

可得$\beta_k = \frac{\boldsymbol{g}^{(k+1)T}[\boldsymbol{g}^{(k+1)} - \boldsymbol{g}^{(k)}]}{\boldsymbol{g}^{(k)T}\boldsymbol{g}^{(k)}}$

非二次型问题

- **Fletcher-Reeves公式：** $\beta_k = \dfrac{\boldsymbol{g}^{(k+1)T}\boldsymbol{g}^{(k+1)}}{\boldsymbol{g}^{(k)T}\boldsymbol{g}^{(k)}}$

通过共轭方向公式

$$\boldsymbol{d}^{(k)} = -\boldsymbol{g}^{(k)} + \beta_{k-1}\boldsymbol{d}^{(k-1)}$$

$\Longrightarrow \boldsymbol{g}^{(k+1)T}\boldsymbol{d}^{(k)} = -\boldsymbol{g}^{(k+1)T}\boldsymbol{g}^{(k)} + \beta_{k-1}\boldsymbol{g}^{(k+1)T}\boldsymbol{d}^{(k-1)}$，

由性质$\boldsymbol{g}^{(k+1)T}\boldsymbol{d}^{(k)} = 0$，$\boldsymbol{g}^{(k+1)T}\boldsymbol{d}^{(k-1)} = 0$得

$\boldsymbol{g}^{(k+1)T}\boldsymbol{g}^{(k)} = 0$，

带入$Polak - Ribière$公式

$$\beta_k = \frac{\boldsymbol{g}^{(k+1)T}\boldsymbol{g}^{(k+1)} - \boldsymbol{g}^{(k+1)T}\boldsymbol{g}^{(k)}}{\boldsymbol{g}^{(k)T}\boldsymbol{g}^{(k)}} = \frac{\boldsymbol{g}^{(k+1)T}\boldsymbol{g}^{(k+1)}}{\boldsymbol{g}^{(k)T}\boldsymbol{g}^{(k)}}$$

非二次型问题

- $\min f(\boldsymbol{x}),\ \boldsymbol{x} \in \mathbb{R}^n$的Fletcher-Reeves共轭梯度算法：

1. $k = 0$，选择初始点$\boldsymbol{x}^{(0)}$，$\boldsymbol{g}^{(0)} = \nabla f\left(x^{(0)}\right)$。如果满足停机条件（如$\left\|\boldsymbol{g}^{(0)}\right\| \leq \varepsilon$），则停止。否则设$\boldsymbol{d}^{(0)} = -\boldsymbol{g}^{(0)}$.
2. 使用一维优化方法计算 $\alpha_k = argmin_\alpha f(\boldsymbol{x}^{(k)} + \alpha \boldsymbol{d}^{(k)}), \boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \alpha_k \boldsymbol{d}^{(k)}, \boldsymbol{g}^{(k+1)} = \nabla f(\boldsymbol{x}^{(k+1)})$,检查停止条件。如满足，返回$\boldsymbol{x}^{(k+1)}$；否则进入下一步.
3. 判断$k = n$是否成立；若是，重置初始点$\boldsymbol{x}^{(0)} = \boldsymbol{x}^{(k+1)}$，$\boldsymbol{g}^{(0)} = \nabla f\left(\boldsymbol{x}^{(0)}\right), \boldsymbol{d}^{(0)} = -\boldsymbol{g}^{(0)}$，$k = 0$返回第二步；若否，进入下一步。
4. 计算$\beta_k = \frac{\boldsymbol{g}^{(k+1)T}\boldsymbol{g}^{(k+1)}}{\boldsymbol{g}^{(k)T}\boldsymbol{g}^{(k)}}, \boldsymbol{d}^{(k+1)} = -\boldsymbol{g}^{(k+1)} + \beta_k \boldsymbol{d}^{(k)}$
5. 令$k = k + 1$,返回第二步

非二次型问题

- 注意：

1. 对于二次型函数，上述三个公式等价。对非二次型函数，3个公式不一致。
2. n次迭代不一定到极小点，一般经过n次迭代后将搜索方向重新初始化为梯度方向。
3. 共轭梯度法的收敛速度是超线性的（证明略）。当$\beta_k = 0$时，共轭梯度法就是最速下降法。

小结

- 二次型共轭方向：$\boldsymbol{d}^{(i)}\boldsymbol{Q}\boldsymbol{d}^{(j)}=0$
- 共轭方向法： 给定共轭方向$\boldsymbol{d}^{(\mathbf{0})},\boldsymbol{d}^{(1)},\boldsymbol{d}^{(2)},\dots,\boldsymbol{d}^{(\mathrm{n}-1)}$

$$\boldsymbol{x}^{(\mathrm{k}+1)}=\boldsymbol{x}^{k}+\alpha_k\boldsymbol{d}^{(k)},\ \alpha_k=-\frac{\boldsymbol{g}^{(k)T}\boldsymbol{d}^{(k)}}{\boldsymbol{d}^{(k)T}\boldsymbol{Q}\boldsymbol{d}^{(k)}}$$

- Gram − Schmidt方法：给定线性无关方向$\boldsymbol{p}^{(0)},\dots,\boldsymbol{p}^{(n-1)}$

$$\boldsymbol{d}^{(k+1)}=\boldsymbol{p}^{(k+1)}-\sum_{i=0}^{k}\frac{\left(\boldsymbol{d}^{(i)}\right)^{T}\boldsymbol{Q}\boldsymbol{p}^{(k+1)}}{(\boldsymbol{d}^{(i)})^{T}\boldsymbol{Q}\boldsymbol{d}^{(i)}}d^{(i)}$$

- 共轭梯度法： $d^{(0)}=-g^{(0)}$， $\boldsymbol{d}^{(k+1)}=-\boldsymbol{g}^{(k+1)}+\beta_k\boldsymbol{d}^{(k)}$其中， $\beta_k=\frac{\left(\boldsymbol{g}^{(k+1)}\right)^{T}\boldsymbol{Q}\boldsymbol{d}^{(k)}}{\left(\boldsymbol{d}^{(k)}\right)^{T}\boldsymbol{Q}\boldsymbol{d}^{(k)}}$
- 一般问题的共轭梯度法：1. 利用一维优化求α_k。2.利用Hestenes-Stiefel公式、 Polak –Ribière公式和Fletcher-Reeves公式求β_k

谢 谢
THANKS